

MUHAMMAD ROSHAAN

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OBJECTIVE

Computer Science 3rd year student with hands-on experience in backend development and data analytics. Skilled in FastAPI, Python, SQL, MySQL/MongoDB, and Power BI. Seeking an internship or junior role to contribute to backend or data-driven projects while gaining industry experience.

EDUCATION

Sir Syed University of Engineering and Technology

BS in Computer Science | 2023 – Present (CGPA:3.81/4.00) (6th Semester)

Courses: Python, Software Engineering, Data Structures, Database Management, OOP, OS

Sep 2023–July 2027

Karachi

TECHNICAL SKILLS

Backend & Architecture: Python (Advanced), FastAPI, Celery, RESTful API Design, SQLAlchemy ORM, WebSockets, JWT Authentication

Concurrency & Infrastructure: Redis (Pub/Sub, Distributed Locks, Caching), Pessimistic Locking, Asynchronous Worker Queues, Git/GitHub

Databases & Cloud: MySQL, PostgreSQL, SQLite, MongoDB, Supabase (Real-Time Sync)

Data Engineering & BI: Pandas, NumPy, Matplotlib, ETL Data Pipelines, Power BI, DAX

Core Competencies: High-Concurrency System Design, Race Condition Elimination, Zero Data-Loss Pipelines, System Stability Under Load

PRACTICAL EXPERIENCE AND PROJECTS

Fleet Command Center (In Progress) / FastAPI + Redis + WebSockets (May 2026 - present)

- Streamed live geospatial data with sub-second latency using Redis Pub/Sub and WebSockets.
- Prevented concurrency race conditions via Redis pessimistic locking and simulated map-matched routing using OSRM.

SeatVault (Concurrency Engine) - Python (FastAPI) + Redis + Celery + React (Mar 2026 – Apr 2026)

- Engineered a high-concurrency ticketing system using Redis distributed locks and FastAPI to prevent double-booking and race conditions during flash sales.
- Integrated Celery for asynchronous background processing and built a React frontend with real-time polling and JWT authentication.

BooknScore - Flutter / Supabase / SQLite (Dec 2025 – Mar 2026)

- Developed an offline-first cricket scoring mobile app with real-time Supabase cloud synchronization.
- Engineered a resilient SQLite scoring engine to prevent data loss during network interruptions.

FlashSale Concurrency Engine – Python (FastAPI) + MySQL (Oct 2025)

- Designed a high-concurrency backend system to eliminate Race Conditions in inventory management.
- Implemented Pessimistic Locking via SQLAlchemy transactions to prevent overselling during simultaneous requests.

LANGUAGES

English: Fluent | Urdu: Native

CERTIFICATIONS AND COURSES

♣ **Introduction to MongoDB in Python** | DataCamp (Nov 2025)

A 3-hour course covering MongoDB fundamentals, NoSQL concepts, and core database operations in Python.

♣ **FastAPI (Python)** | Udemy | July - Oct 2025

Learned to build and deploy high-performance RESTful APIs with FastAPI, including database integration, data validation, and modular application design (21.5 hours).